

Texas Holdem Poker

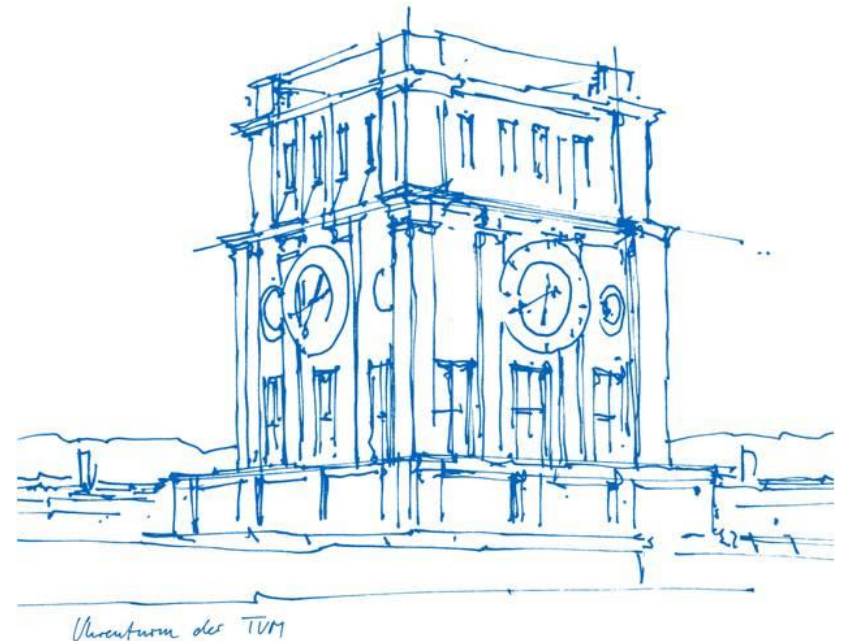
Technische Universität München

TUM School of Management

Programming Texas Holdem

Info for review: Names on Last Slide

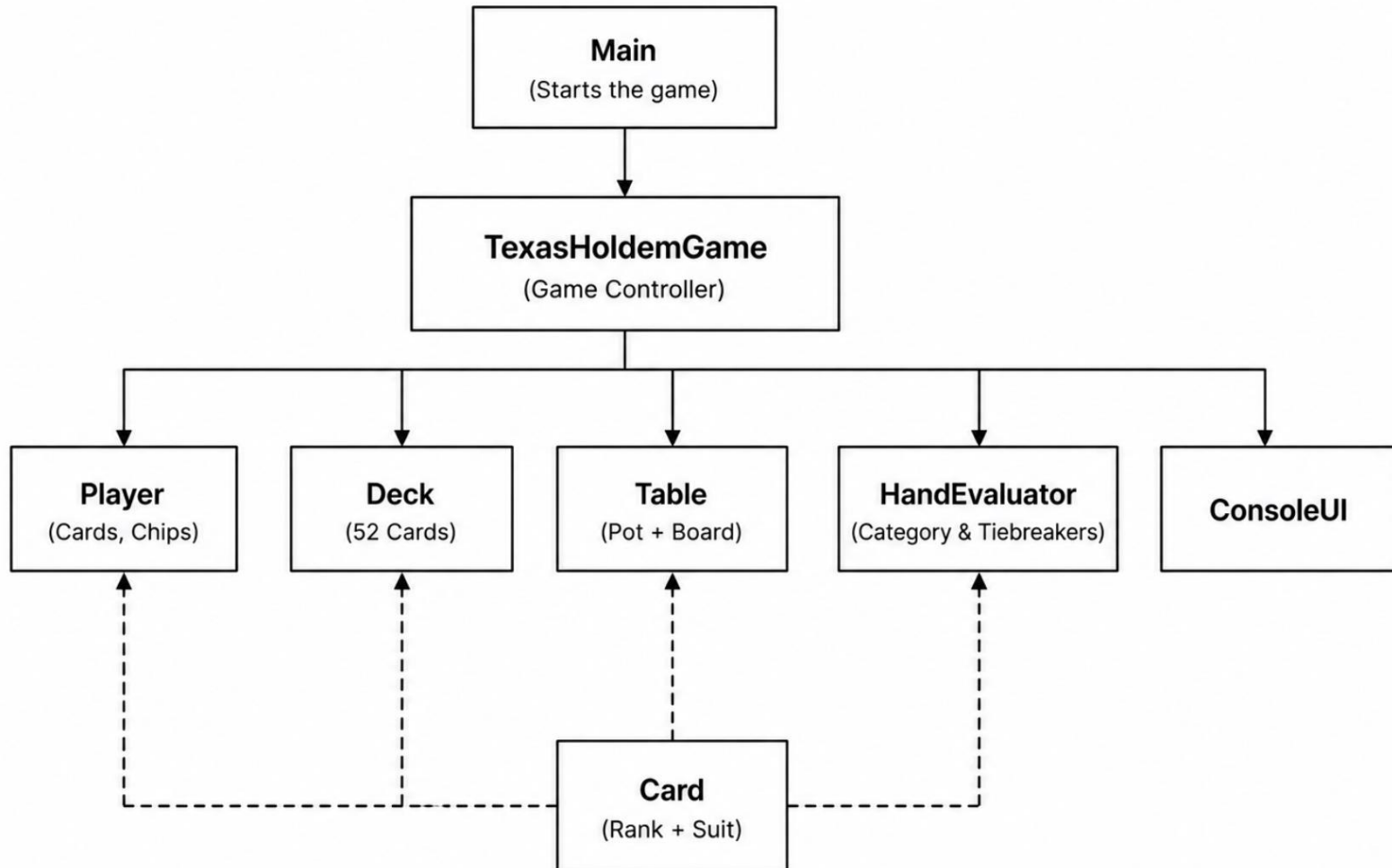
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Process

Provided Classes	Web-Interface	Win Probability per Move
		Evaluation • Matplotlib (local) • Dynamic online visualization
Completed and refined provided classes	Designed and implemented GUI	Challenge Starting wrong approach
		Decision Dynamic online integration Monte carlo

Architecture



Monte Carlo

```
def monteCarlo(self, hole_cards, community_cards, num_opponents, simulations=1000):
    full_deck = [Card(rank, suit) for suit in Suit for rank in Rank]
    wins = 0
    for _ in range(simulations):
        known = set(hole_cards + community_cards)
        remaining = [c for c in full_deck if c not in known]
        shuffle(remaining)

        cards_needed = 5 - len(community_cards)
        board = community_cards + remaining[:cards_needed]
        remaining = remaining[cards_needed:]

        my_rank = self.best_rank(hole_cards + board)
        opponent_ranks = [
            self.best_rank(remaining[i*2:(i+1)*2] + board)
            for i in range(num_opponents)
        ]

        if my_rank >= max(opponent_ranks):
            wins += 1

    return wins / simulations
```

Future Improvement

Switching Blinds & Betting Agents

Rework the current logic

Multi-user & Multithreading

Every new user gets allocated to a new instance

For Review: AI was used for extra Tasks!

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